

Abhishek Meghwal

Machine Learning Engineer | Data Analyst

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PROFILE SUMMARY

- Results-driven **Data Scientist with hands on experience** in demand forecasting, predictive analytics, and machine learning within energy domain.
- Delivering high impact forecasting solutions at **ExxonMobil**, spearheading initiatives that optimize operations and reduce working capital through advanced analytics and forecasting, feature engineering and data validation.
- Proficient in **Python, SQL, and Pyspark**, with a proven ability to convert complex data into actionable insights, streamline ML pipelines, and drive strategic business decisions.
- Specialized in **Deep Learning and Computer Vision**, developing models for image-to-image translation and automated defect detection in industrial applications.



CORE COMPETENCIES

- Demand Forecasting & Predictive Analytics
- Time Series Analysis & Feature Engineering
- Anomaly Detection & Quality Assurance
- Machine Learning & Deep Learning
- ML Pipeline Development & Automation
- Image Processing & Computer Vision
- Data Exploration & Statistical Modeling
- Model Optimization & Hyperparameter Tuning
- Data Engineering & ETL Management



WORK EXPERIENCE

ExxonMobil Company India Pvt. Ltd., Bangalore | Data Scientist | 2024-07 to 2025-10

ExxonMobil

- Pioneered the advanced demand forecasting solution, delivering **\$17M annual savings** in working capital by integrating predictive modeling with business-driven strategies.
- Improved forecast **accuracy by 1.64%** and mitigated bias through feature engineering, seasonality modeling, and hyperparameter tuning.
- Diagnosed and resolved critical data anomalies, **enhancing back-testing accuracy by 1.16%** and strengthening enterprise-wide decision confidence.
- Built regional accuracy & bias assessment frameworks, automating monthly performance reports and reducing turnaround time by **80% (6hrs to ~1hr)**.
- Led pipeline enhancements, **reduced runtime by ~50%** and developed a forecast QA pipeline which detected early pipeline issues, thereby improving model reliability and **dev cycle efficiency by 30%**.
- Collaborated with **3 cross-functional teams** (Data Engg., MLE, Business) to automate the data delivery pipelines reducing manual intervention by **~40%**.
- Conducted in-depth exploratory data analysis for **3 regions and 52K SKUs**, uncovering holidays & events, lifting patterns which directly informed the model improvement strategies.



PROJECTS UNDERTAKEN

RGB to Thermal | Image-to-Image Translation | 2024-02 | [Link](#)

- Implemented **Pix2Pix GAN model** for image translation, leveraging **U-Net generator** and **Patch GAN discriminator**, training on 1,000 paired images.
- Achieved **PSNR (Peak Signal to Noise Ratio) of 27.19** on validation data, demonstrating high-fidelity image translation.
- Enhanced model performance through data augmentation, multi-loss optimization, and tuning adversarial-to-L1 loss ratios.

Metal Surface Defects Classification using ResNet18 | 2023-11 | [Link](#)

- Engineered modular, production ready **PyTorch** pipeline on NEU dataset of **1,800 images**, improving reliability of inspection process.
- Implemented **transfer learning**, model checkpointing, early stopping and data augmentation achieving **97.4% train and 98.6% test accuracy**.
- Built a scalable, **config-driven** pipeline with **unit tests**, reproducible runs, and automated predictions following MLOps best practices.

Loan Approval Prediction | 2023-06 | [Link](#)

- Implemented an end-to-end ML pipeline, covering data cleaning, preprocessing, feature engineering, EDA, and model evaluation on **614** applicants.
- Trained and compared **7 ML classifiers** identifying KNN as the top-performing model with **80.54% accuracy** and an **F1-score of 0.80**.
- Improved model performance through hyperparameter tuning, selecting **K=1** as the optimal with **~2%** accuracy lift over baseline Random Forest.

HR Analytics Dashboard | 2023-11 | [Link](#)

- Built an interactive HR Analytics Dashboard in Power BI, analyzing employee attrition trends across departments and job roles for **1,470** employees.
- Identified key attrition drivers such as **16.1% attrition rates**, highest exit in **R&D (133 employees)**, salary bands **below 5K**, early tenure employees.
- Transformed HR datasets into actionable insights, highlighting role-specific attrition and age trends (**26-35**), enabling data-backed decision making.



TECHNICAL SKILLS

- Languages:** Python, C++, SQL
- ML & Analytics:** Pandas, Pyspark, Scikit-Learn, TensorFlow, Keras, Kedro, Time Series Forecasting, EDA, Matplotlib
- Tools & Frameworks:** Snowflake, Databricks, Azure Data Factory, VS-Code, GitHub, Agile (SAFe Certified)



EDUCATION & CERTIFICATIONS

- IIT Madras | M.Tech., in Manufacturing Engineering | CGPA: 9.36 | May 2024**
- DTU, Delhi | B.Tech. in Mechanical Engineering | CGPA: 7.9 | Aug 2021**
- SAFe Certified Practitioner** – Agile fundamentals & framework
- Coursera & LinkedIn:** Deep Learning for Imaging, Intro to ML, Probability & Statistics for ML, Git Essentials, Supply Chain Foundations



LEADERSHIP & EXTRA EXTRACURRICULAR ACTIVITIES

- Recruitment Team – ExxonMobil | Campus Ambassador | Jan'25-Oct'25
- Placement Team – IIT Madras | Placement Coordinator | Aug'22 – Jul'24
- Mechanical Department – IIT Madras | Class Representative | Aug'22 – Jul'24
- Mechanical Department – IIT Madras | Teaching Assistant | Jan'24 – Present